Standalone Smart Hajj Earbud Assistant

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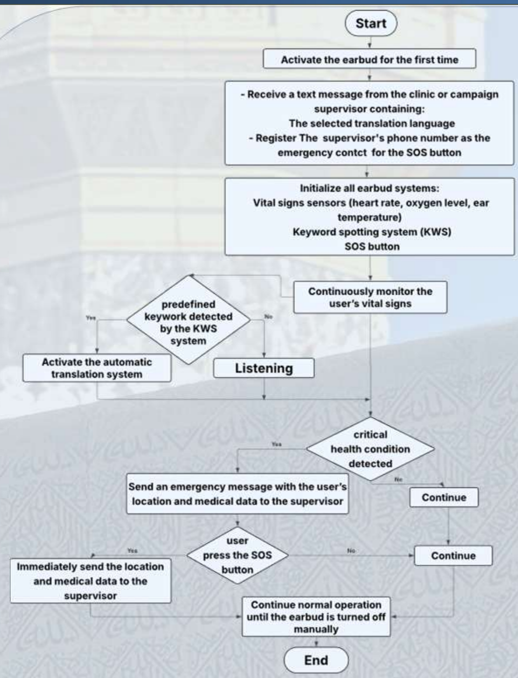
Introduction

Hajj is one of the largest religious gatherings in the world, millions of pilgrims face significant health, physical, and communication challenges . To address these challenges, this project proposes a standalone smart earbud that integrates translation, vital signs monitoring, and location tracking. enables effective communication between pilgrims and medical teams. In emergency situations, it automatically transmits the pilgrim's location and health data via cellular networks for rapid intervention. This solution enhances safety, supports Saudi Vision 2030 by leveraging digital technologies to provide a safer and more efficient Hajj experience.

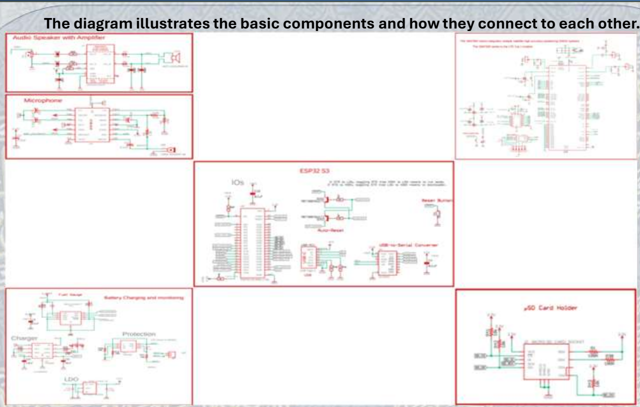
System model



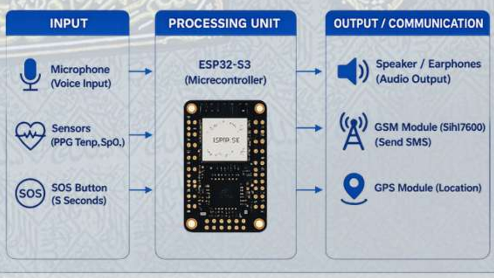
The proposed system provides the following functionalities:

System Block Diagrams

The diagram illustrates the basic components and how they connect to each other.

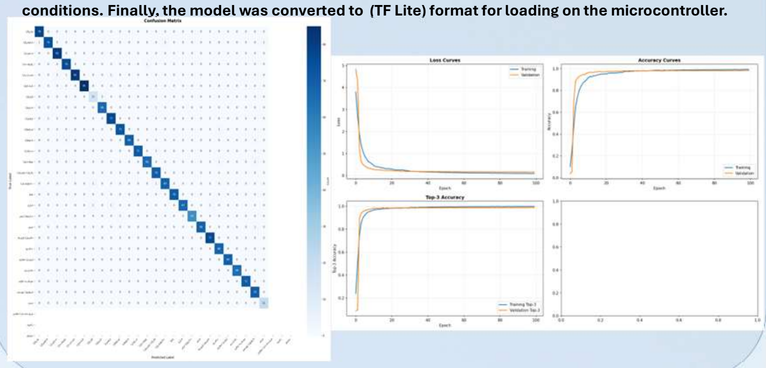


Block diagram illustrating the system architecture, including input, processing, and output :



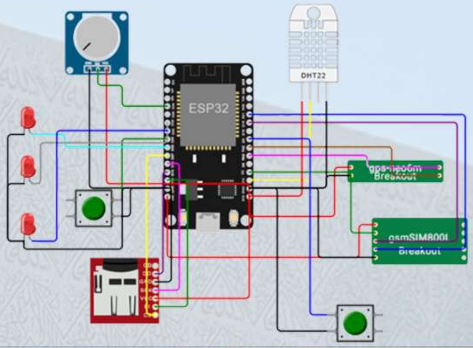
Software Specifications

The model was developed using Google Collab with GPU support to enhance training efficiency. Audio preprocessing techniques such as noise and echo were applied to simulate real-world conditions. Finally, the model was converted to (TF Lite) format for loading on the microcontroller.



Simulations

The circuit was simulated to evaluate system performance . The results confirmed successful system integration and reliable response, demonstrating readiness for real-world implementation.



Proposed Design Views

The following views represent the proposed design of the device :

Conclusions

this project is about developing an integrated in-ear health monitoring system. focuses on utilizing the benefits of the ear canal as an ideal location for accurate monitoring and comfortable experience. The significance of this project is its ability to enhance individual healthcare. By allowing ongoing measurements, offering translation and ability to ask for help from anywhere. This project contributes to digital transformation for a better hajj experience.

Selected References

- *Mass Gathering Medicine*, "Lessons from tragedy: Hajj deaths," 2024.
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- L. Goyal et al., "Language Translator Headphone Design," *E3S Web Conf.*, 2023.
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